Radar target classification from Doppler observations

Abstract. We investigate nine-class radar target classification from short complex I/Q observations. Doppler power spectra motivate interpretable features and a compact CNN. The selected peak-aligned CNN with whole-piece probability averaging achieves **0.866 mean-fold Macro-F1**, averaged over three robustness assignments, compared with **0.917** on the development assignment used for model selection. Both evaluations hold out complete recordings. We emphasize the robustness average as the more cautious internal performance estimate. The assignments reuse the same 87 recordings; they are not independent test sets. Limited recording diversity and unresolved signal-quality confounding constrain expectations for unseen conditions.

1. Measurements: complex I/Q to Doppler power

Each observation contains 256 complex samples from each of five range cells measured by a 77 GHz FMCW radar [1,2]. Range resolution is approximately 1 m; the target is centered in cell 2.

Table 1. Supplied subset. Segments are correlated within recordings; IDs do not establish independent targets or sessions.

Class	Target	Segments	Recordings
D1	DJI Matrice 200 V2	5,350	6
D2	DJI Mavic 2 Pro	7,364	4
D3	DJI Phantom 3	8,807	3
D4	Custom FPV drone	7,778	3
D5	Tarot 680 Pro frame	9,531	3
D6	Syma X23W	8,788	11
human	Walking or running	5,205	7
bird	Multiple bird species	6,262	36
reflector	Stationary corner reflector	2,588	14
Total		61,673	87

Chirps repeat at 17 kHz (PRF), giving 15.06 ms per segment; observations recur about 100 ms apart. Phase evolution across chirps contains Doppler information. Equations (1)-(3) define complex I/Q, its untapered FFT and real-valued Doppler power:

$$x_r[n] = I_r[n] + jQ_r[n] \tag{1}$$

$$X_r[k] = \sum_{n=0}^{255} x_r[n]\exp(-j\,2\pi kn/256) \tag{2}$$

$$P_r[k] = |\mathrm{fftshift}(X_r)[k]|^2/256^2 \tag{3}$$

Here r=0…4 indexes cells; n,k=0…255 index chirps and frequency bins. I and Q are real and imaginary components, j^2=-1, X is the complex DFT and P its Doppler power. No taper is applied. FFT shift centers zero frequency.

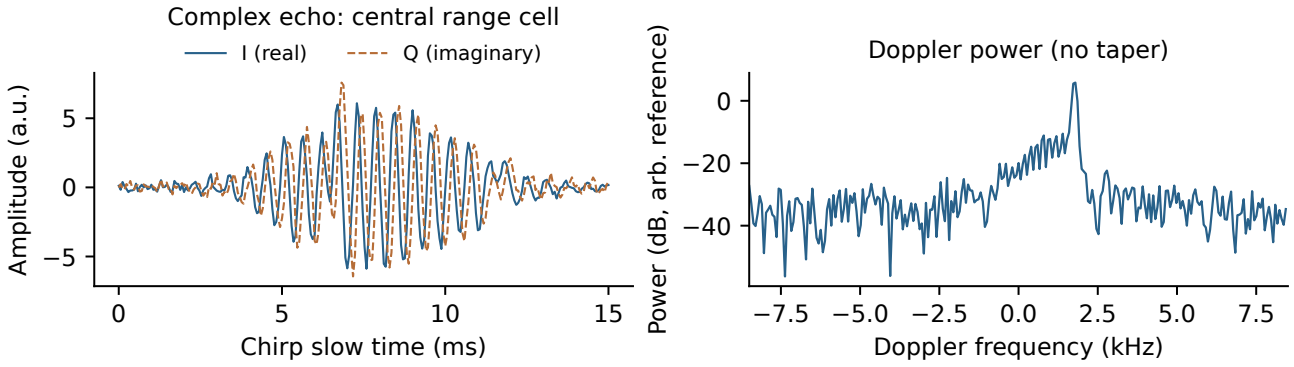


Figure 1. D1 recording 6, row 5,089, cell 2: overlaid I/Q (left) and the same observation's Doppler power (right). Power is scaled by 256^2 with an arbitrary dB reference. Chirp slow time is not time between scans.

2. Spectral diversity and physically interpretable summaries

Physical interpretation. D1-D3 peaks lie near zero; D4-D6 peak positions and surrounding power vary. Translation shifts Doppler; rotating blades can add components, without uniquely identifying scatterers. Human clusters are consistent with torso/limb motion. Bird examples differ in peak structure; short segments need not capture a wingbeat. Reflector examples concentrate near zero, consistent with little radial motion and no internal moving parts. Aspect, clutter and noise also affect spectra; these are not universal class signatures.

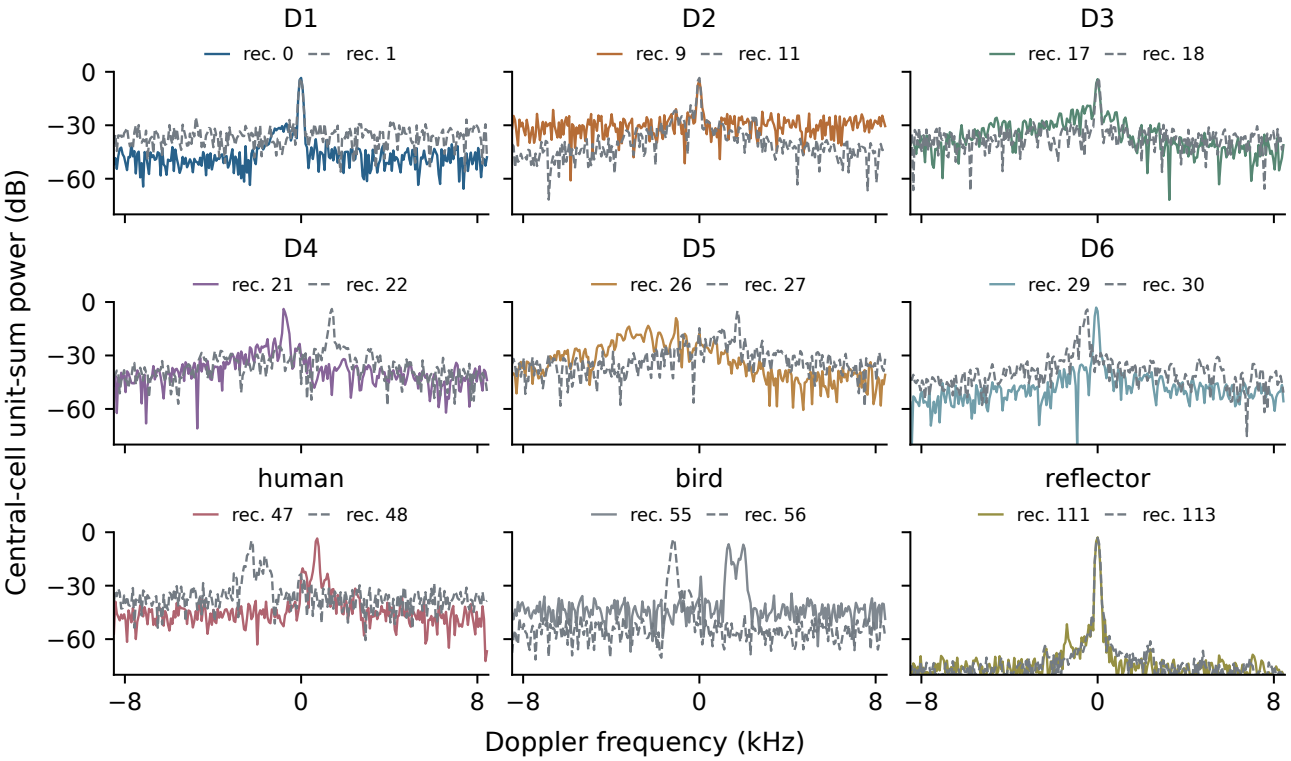


Figure 2. Two recording-midpoint central-cell spectra per class, with unit-sum power before dB conversion. Shared axes and solid/dashed traces show within-class variation and between-class overlap.

The 256-chirp FFT gives 66.4 Hz bins over approximately ±8.5 kHz. DC is retained; the untapered FFT can exhibit boundary leakage. Equation (4) normalizes central-cell power; Eqs. (5)-(11) define five descriptors:

$$p_k = \frac{P_2[k]}{\sum_{\ell} P_2[\ell]} \quad (4) \qquad k_* = \arg\max_k p_k \quad (5)$$

$$f_* = f_{k_*} \quad (6) \qquad \mu_f = \sum_k p_k f_k \quad (7)$$

$$\sigma_f = \sqrt{\sum_k p_k (f_k - \mu_f)^2} \quad (8) \qquad H_f = -\frac{\sum_k p_k \log p_k}{\log 256} \quad (9)$$

$$C = \frac{p_{k_*}}{\mathrm{median}_{|f_k - f_*| > 250\,\mathrm{Hz}}\, p_k} \quad (10) \qquad F_{\mathrm{off}} = \sum_{|f_k - f_*| > 250\,\mathrm{Hz}} p_k \quad (11)$$

Here p is normalized linear power, f is frequency in Hz, and the starred index denotes the dominant bin. Numerical floors handle zero power.

Peak frequency (Eq. 6) locates the dominant return, not necessarily body motion. **Bandwidth** (Eq. 8) is power-weighted standard deviation; **entropy** (Eq. 9), spectral dispersion. **Contrast** (Eq. 10) compares peak power with the median outside ±250 Hz; **off-peak fraction** (Eq. 11) sums power there. Contrast is not calibrated SNR.

5→9: finer central-cell shape. Add asymmetry (power above minus below the peak), secondary/main-peak ratio (strongest local peak at least four bins away), and near/far fractions at 3-8 and >8 bins. These describe imbalance, multiple components and energy spread. All nine use the unaligned central spectrum; these boundaries differ from the ±250 Hz exclusion.

3. From cell relationships to frequency-dependent energy

9→17: whole-spectrum cell relationships. Four fractions measure each cell's power relative to the five-cell total; cell 4 is omitted because it is implied. Four cosine similarities (normalized dot products) compare unaligned central and neighbor spectra (cells 0,1,3,4). Values near one indicate similar shapes; near zero, little overlap. Together these test whether energy distribution and shared Doppler structure aid classification.

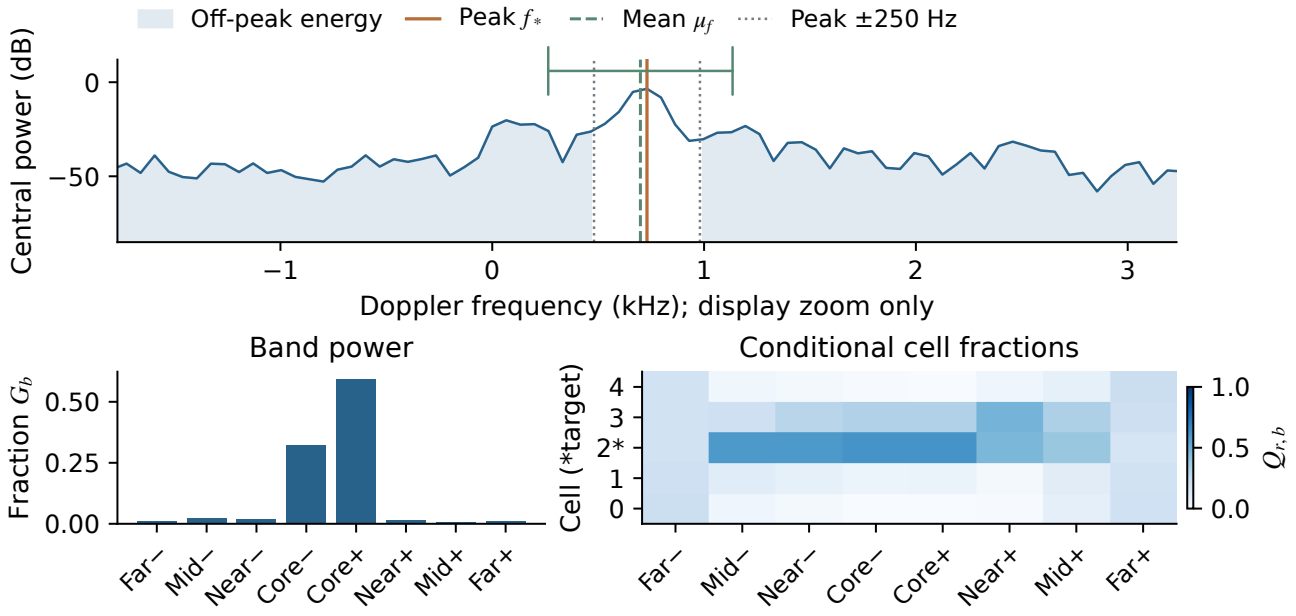


Figure 3. Human recording 47, row 48,131 (recording midpoint). Top: peak, mean, ±250 Hz exclusion and off-peak shading; the green bracket is μ±σ from **all 256 bins**, not a threshold width. Bottom: band power and conditional cell fractions. Gray cells indicate unavailable values.

17→65: global features G1-G3. Global means all five cells within one segment. To resolve frequency-dependent cell structure, we split spectra into eight peak-relative bands. Equation (12) gives each cell's band power E. Bands use FFT-bin offsets, not the original ±250 Hz exclusion:

$$E_{r,b} = \sum_{k \in b} P_r[k] \qquad (12)$$

$$G_b = \frac{\sum_r E_{r,b}}{\sum_s \sum_k P_s[k]} \qquad (13)$$

$$Q_{r,b} = \frac{E_{r,b}}{\sum_s E_{s,b}} \qquad (14)$$

$$H_b = -\frac{\sum_r Q_{r,b} \log Q_{r,b}}{\log 5} \qquad (15)$$

G1—band power (17→25): Eq. (13) measures each band's fraction of total five-cell power (8 features). **G2—cell distribution (25→57):** Eq. (14) measures cell distribution within each band; keep four fractions because the fifth is implied (8×4=32). **G3—spatial entropy (57→65):** Eq. (15) measures concentration, from one-cell power (0) to equal power (1), adding eight nonlinear summaries.

Table 2. Development **pooled OOF Macro-F1** on identical recording folds. Rows show cumulative feature totals. Independent-segment results do not match the inference context of whole-piece averaging.

Total	Added features	LR	XGBoost
5	5 central-cell spectral summaries	0.3887	0.5853
9	+4: asymmetry, secondary-peak ratio, near/far fractions	0.4838	0.6363
17	+4 cell-power fractions; +4 cosine similarities	0.5408	0.6737
25	G1: +8 global band-power fractions	0.5996	0.7148
57	G2: +8 bands x 4 within-band cell fractions	0.7154	0.7237
65	G3: +8 within-band spatial entropies	0.7430	0.7386

We fix LR at L2 regularization with C=1 and XGBoost at 100 depth-three trees, learning rate 0.1 and L2=1. Missing features are imputed, and LR scaling fitted, using training data only. Engineered features use linear power; CNN inputs use normalized log power. CNN comparisons with averaging change context, not architecture alone.

4. CNN, peak alignment and whole-piece probability averaging

A 7,785-parameter 1D CNN retains spectral detail lost by summaries. Its frequency-axis convolutions learn local patterns while combining all five range cells.

Table 3. CNN architecture.

Layer	Specification
Input	5 cells x 256 frequency bins
Conv/ReLU/pool	5->16, k7/p3; max-pool2
Conv/ReLU	16->32, k5/p2
Adaptive pool	16 positions; flatten 512
Class head	Linear512->9; softmax

Table 4. Development **mean-fold Macro-F1**, fold SD.

CNN input	Mean-fold F1	Fold SD
Original	0.7692	0.0411
Peak aligned	0.8182	0.0164

Peak alignment centers the **central-cell maximum**, whereas FFT shift centers zero frequency. All five spectra shift equally without wrap. We normalize retained five-cell power, apply 10 $\log_{10}$, floor at -80 dB and divide by 80.

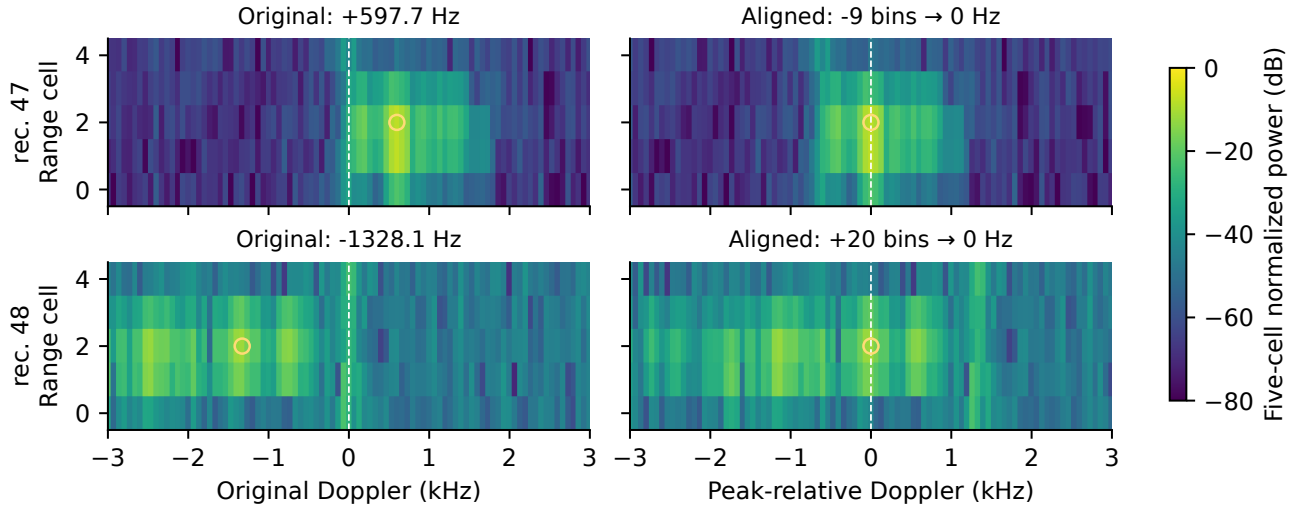
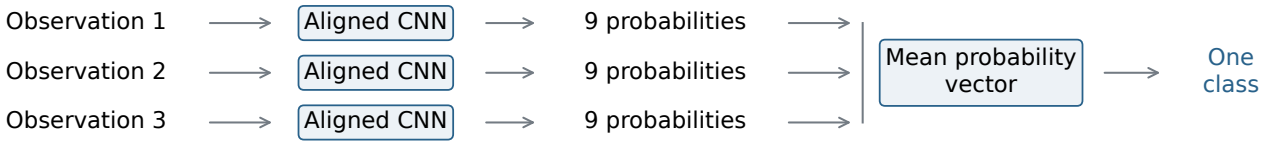


Figure 4. Human recording 47 / row 47,661 and recording 48 / row 48,883. Labels show measured untapered peak frequencies and shifts. Display: ±3 kHz; processing: 256 bins. Shifts truncate/zero-pad edges. Peaks need not indicate body velocity; resolution/SNR do not increase.

Choices tested and dropped. We retained peak alignment and whole-piece averaging after matched recording-held-out comparisons. Unaligned inputs performed worse. Hann tapering gave similar average scores with mixed recording effects; rectangular was retained for simplicity, not demonstrated superiority. Both CNNs in Table 4 use the untapered FFT. Table 5 compares independent, local-five and whole-piece inference for all models. Averaging adds no learned parameters. For T=1…30 observations, p denotes class-c probability at observation t:



Equation (16) averages **class probabilities**; Eq. (17) assigns one class to the piece, assuming a shared target.

$$\bar{p}_c = \frac{1}{T}\sum_{t=1}^{T} p_{t,c} \qquad (16)$$

$$\hat{y} = \arg\max_c \bar{p}_c \qquad (17)$$

Table 5. Development **pooled OOF Macro-F1**, matched held-out data. LR/XGBoost: 65 features. Local-five uses centered, truncated windows; whole-piece averages at most 30 observations.

Model / input	Independent	Local-five	Whole-piece
LR / 65 features	0.7430	0.8021	0.8308
XGBoost / 65 features	0.7386	0.7896	0.8275
CNN / aligned spectra	0.8205	0.8835	0.9175

5. Recording-held-out evaluation and failures

Development. Correlated segments make random splits optimistic. Class-aware three-fold cross-validation (seed 42) holds out complete recordings, approximately balancing within-class segment counts. Each recording validates once; every fold covers all nine classes. Models start fresh; learned preprocessing uses training data only.

Robustness. Seeds 7,21,84 reassign the same 87 recordings, measuring fold sensitivity rather than independent test performance. Network initialization/shuffling remains seed 42. CNN mean-fold F1 (independent/local-five/whole-piece): 0.8182/0.8821/0.9170 for development; 0.7877/0.8352/0.8656 averaged over robustness assignments.

D3/D4/D5 each have only three recordings. Each held-out D3 recording leaves the same two D3 training recordings; repeated assignments change other classes' allocation, not D3 training diversity.

Metrics. Accuracy weights segments; Macro-F1 averages class F1 scores. Mean-fold Macro-F1 averages fold scores, while pooled OOF Macro-F1 scores all held-out predictions together. Class-balanced recording accuracy averages recording accuracies within classes, then across classes. Development pooled OOF Macro-F1 is **0.9175**, accuracy **0.9099**, and class-balanced recording accuracy **0.8731**, from fold models rather than the final full-data checkpoint.

CNN training uses 20 epochs, Adam (learning rate 0.001, weight decay 0.0001), seed 42 and batch size 256; no outer-validation stopping. Cross-entropy is unweighted, without balanced sampling; class weighting was not tested. Recording-wise pieces split gaps >0.5 s and contain at most 30 segments; inference uses supplied IDs. Range/near-edge inputs are unused.

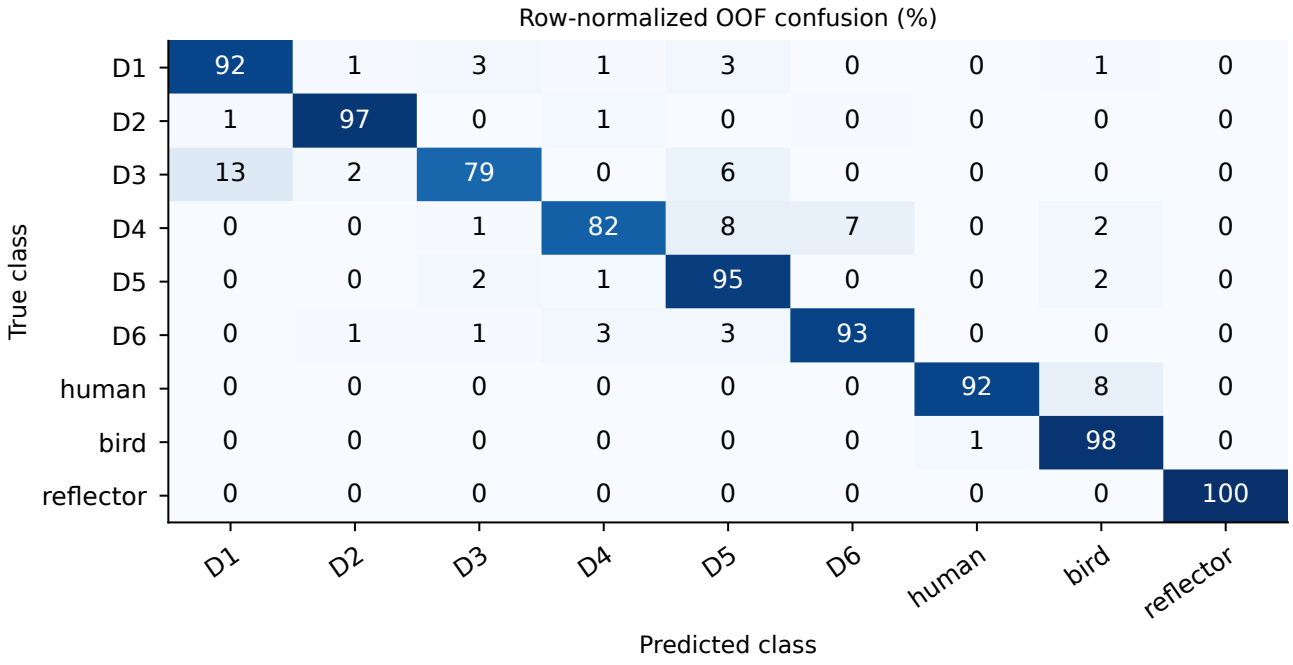


Figure 5. Aligned CNN with whole-piece probability averaging: row-normalized pooled OOF confusion percentages; the diagonal gives class recall. Each of 61,673 observations appears once. Largest error: D3 to D1 (1,119 observations). Strong class recall can hide failed small recordings.

Table 6. Confirmed difficult recordings for the selected pipeline.

Recording	Class	Segments	Accuracy
9	D2	98	0.000
18	D3	2469	0.547
32	D6	557	0.431
68	bird	19	0.000

Range and failures. D2/9 (0% accuracy) and D3/18 (54.7%) lie entirely beyond their same-class training ranges: 78.3-89.6 versus 22.6-47.2 m, and 57.5-59.4 versus 15.1-29.2 m. D6/32 (43.1%) has median range 41.5 m versus training-recording medians 13.2-20.8 m, but ranges overlap. Conversely, D1/3 achieves 99.9% beyond its same-class training range. Distance therefore accompanies some failures without establishing their cause; motion, aspect, clutter and signal quality remain possible factors.

6. Signal quality, uncertainty and conclusions

The assignment warns that signal strength can become a shortcut. Total-power normalization removes a common amplitude scale but retains spectral contrast, noise-floor shape and relative power distribution. Peak/background contrast is an explicit classical feature and is available in the CNN input; we have not established whether the CNN uses it.

Held-out accuracy across peak/background-contrast quintiles was 0.888, 0.870, 0.910, 0.923, 0.959. This proxy excludes ±250 Hz around the original peak but is not calibrated SNR: background includes noise, clutter, leakage and target returns. Changing class/recording composition prevents causal interpretation; no monotonic trend does not establish SNR independence.

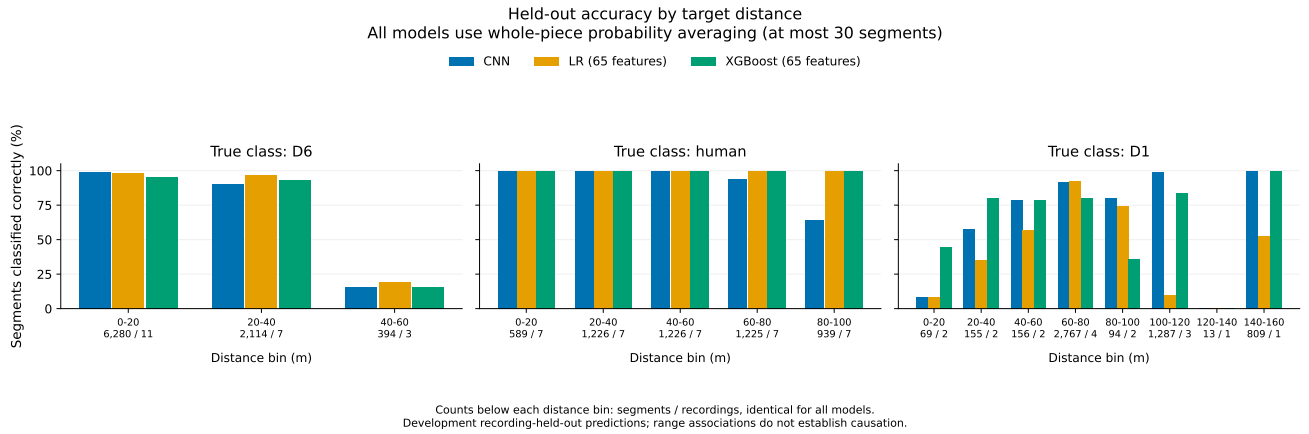


Figure 6. Range-binned accuracy for three illustrative true classes, using identical development held-out segments and whole-piece averaging. LR/XGBoost use 65 features. Counts are all segments / contributing recordings, not correct predictions. Each segment uses its own range; pieces and recordings may span bins. Observations are correlated.

Shared and model-specific failures. At 40-60 m, D6 accuracy falls to 15.5%/19.5%/15.5% for CNN/LR/XGBoost. For humans at 80-100 m, CNN accuracy is 63.9%, versus 99.9%/100% for LR/XGBoost; all seven human recordings contribute. D1 shows no consistent decline: at 140-160 m CNN/XGBoost reach 99.9%. Its 0% bin at 120-140 m contains only 13 segments from one recording. Range effects therefore vary by class and model; motion, aspect and signal quality remain confounded.

Recording diversity limits generalization claims. The 87 recordings are unevenly distributed across classes, with only three each for D3/D4/D5. Their independence in objects and acquisition sessions is unknown. Some failures occur beyond same-class training distances, so generalization across range is not established. Excluding range as an input does not remove range-related signal differences. Correlated segments and repeated assignments cannot replace diverse recordings; exploratory reuse also limits confidence. Hidden-recording performance remains unknown.

Motion and representation. D1 recordings 5/6 show substantial range variation and nonzero Doppler, supporting radial motion. Whole-piece-averaged segment accuracy is 87.7%/38.2%, showing uneven recognition across moving-target recordings, rather than complete failure on moving D1 targets. Alignment may follow non-body scatterers, discard energy and introduce padding cues. Neighbor-cell energy may be clutter; association errors can spread across a piece. These results do not establish target-specific micro-Doppler learning. Rare noise-filled near-edge observations have not been shown to explain failures.

Next steps (untested). Collect classes at overlapping distances and varied motion/aspects; evaluate on independent recordings. First inspect errors against range within existing recordings and test controlled noise on matched held-out signals. If supported, test training-only noise augmentation; it cannot reproduce all distance effects. Tuning on known failures risks validation overfitting.

Conclusion. The peak-aligned CNN with whole-piece probability averaging achieves **0.866 mean-fold Macro-F1** across robustness assignments, versus **0.917** on the development assignment used for model selection. The robustness average is our headline internal estimate. Engineered features improve classical baselines; the CNN retains finer spectral detail, and averaging adds gains without learned parameters. Reused recordings do not establish independent hidden-test performance.

References. [1] Supplied *Home project: what is that target?*, sections 2-6. [2] A. Karlsson, *Radar measurements on drones, birds and humans with a 77GHz FMCW sensor*, KTH/Zenodo, doi:10.5281/zenodo.5845259, CC BY 4.0. Only the supplied reshaped nine-class subset was used; no additional radar recordings were obtained.